Beyond Implicit Bias: The Insignificance of SGD Noise in Online Learning

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Date: 2024

Motivation: Online Learning Context

- Online learning (single epoch) dominates LLM training (e.g., GPT-2, Llama)
- Prior work assumes SGD noise provides implicit bias advantages universally
- Critical question: Does SGD noise matter in online learning?

Golden Path Hypothesis

- Proposed: SGD path $\approx$ noiseless GD path in online learning
- Contrast with "Fork in the Road" (offline: noise changes final model)
- Figure 2: Illustrates both scenarios with key implications

Offline vs Online Learning Contrast

- **Offline learning (multi-epoch)**:
  - Smaller batch sizes → better minima (flatter, generalize better)
  - Figure 1 (left): Lower batch size plateaus at lower loss
- **Online learning (single epoch)**:
  - Larger batch sizes → better performance (no implicit bias)
  - Figure 1 (right): Higher batch size reaches lower loss

Theoretical Support (Convex Case)

- **Equality of paths (Theorem 2.1):** In the online convex/quadratic setting, the expected SGD iterate equals the noiseless GD iterate at every step: E[w_SGD(t)] = w_GD(t). SGD noise only adds variance around the same trajectory.
- **Formalizes the golden path hypothesis:** In the convex case, noise introduces no new implicit bias in expectation.
- **Implication:** Batch size in online convex training affects optimization noise and compute — not the learned function's implicit bias.

Snapping Back to Golden Path

- Experiment: Reduce noise at T_0 steps (increase batch size)
- Result (Figure 4):
  - Loss sharply improves → follows low-noise trajectory
  - "Snaps" to golden path within $\tau \ll T_0$ steps
- Contrasts with offline learning (no snapping at convergence)

Function Space Evidence

- Measured total variation (TV) distance between models
- Golden path prediction:
  - After noise reduction, TV $\approx$ two low-noise models
  - Not $\approx$ high-noise vs low-noise models
- Figure 6: Confirms TV distance matches baseline

Implications & Conclusion

- **Practitioners**: Batch size choice driven by computation, not implicit bias
- **Theorists**: Focus noise effects on offline learning regime
- Takeaway: SGD in online learning = noisy GD steps on "golden path"